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- For a single word produced by particular grammar if its derivation have more than one syntax tree then CFG is **ambiguous**.
  - $S \rightarrow aSa \mid bSb \mid a \mid b$ 
    - This is unambiguous CFG
  - $S \rightarrow Sa \mid aS \mid a$ 
    - This is ambiguous CFG. (e.g aa has two different tree syntax possible)
  - One example enough to prove that a CFG is ambiguous.
  - $S \rightarrow aSb \mid Sb \mid Sa \mid a$ 
    - This is ambiguous CFG. (e.g. aab has two different tree syntax possible)
  - $S \rightarrow XaX$
  - $X \rightarrow aX \mid bX \mid \Lambda$ 
    - This is ambiguous CFG. (e.g. aa has two different tree syntax possible)
- Total Language Tree**
  - Use left most substitution first.
  - Entire tree of all possible words from a CFG.
- Push Down Automata (PDA)**
  - Finite Automata can be made of regular languages but not irregular languages.
  - PDA can be used to create regular and irregular languages.
  - In PDA, we have the ability to memorize previous inputs using stack.
  - Pumping lemma is used to differentiate between regular and irregular languages.
  - We need D (letter of a language), T (stack characters), input tape (consists of numbered cells), stack (with a blank character), oval symbols for START, ACCEPT, REJECT, diamond symbols of READ, POP, rectangle symbol for PUSH.
- PDA of words having double a

